Unit Cat1-CN

SKU:U204

























Description

Unit Cat1-CN is an LTE Cat.1 wireless communication unit equipped with the ML307R-DL module. It supports B1 / B3 / B5 / B8 / B34 / B38 / B39 / B40 / B41 bands, compatible with major Chinese network operators. This unit uses a UART communication interface and is controlled via AT commands, supporting TCP/IP, SSL, HTTP, and MQTT protocols. It features an onboard SMA antenna interface and Nano-SIM card slot, suitable for smart home, industrial control, healthcare, and environmental monitoring IoT applications.

Features

- Based on ML307R-DL module solution
- Supports up to 6 concurrent socket connections
- Supports multiple network protocols: TCP/IP, SSL, HTTP, MQTT, etc.
- Supports multiple mainstream frequency bands:
  - LTE-FDD: B1 / B3 / B5 / B8
  - LTE-TDD: B34 / B38 / B39 / B40 / B41
- UART communication interface

Includes

- 1 x Unit Cat1-CN
- 1 x HY2.0-4P Grove Cable (20cm)
- 1 x 850 ~ 2700 MHz SMA Rubber Antenna
- 1 x Hex Key L-Shape 1.5mm (For M2 Screw)

Applications

- Smart Home
- Industrial Control
- Healthcare
- Environmental Monitoring

Specifications

Specification	Parameter
Communication Module	ML307R-DL
Theoretical Speed	LTE-FDD: Max downlink 10Mbps, Max uplink 5Mbps LTE-TDD: Max downlink 9.010Mbps, Max uplink 3.096Mbps
Operating Bands	LTE-FDD: B1 / B3 / B5 / B8 LTE-TDD: B34 / B38 / B39 / B40 / B41
Network Protocols	IPv4 / IPv6 / PING / NTP / DNS / TCP / UDP / HTTP / HTTPS / MQTT / MQTTS
3GPP Standard	R13
Supply Voltage	5V
Communication Method	Default: UART 115200 bps@8N1, modifiable
Operating Power Consumption	TCP connection: TX mode (instant data): CMCC, FDD, B39: DC 5V@171.91mA; CMCC, TDD, B3: DC 5V@831.12mA Standby mode: DC 5V@9.96mA Sleep mode: DC 5V@1.48mA
Antenna Specification	Size 108 x 10mm, Interface Type: SMA (Male), Frequency Band: 850 ~ 2700 MHz, Gain: 3dBi, VSWR ≤ 1.5
Product Size	71.4 x 24.0 x 8.0mm
Product Weight	13.1g
Package Size	138.0 x 93.0 x 10.0mm
Gross Weight	26.4g

Learn

Before using Unit Cat1-CN, please make sure that a Nano-SIM card supporting LTE Cat1 network (from China Mobile, China Unicom, China Telecom, etc.) has been inserted and that data service is activated. The Nano-SIM card slot is located inside the module. Use a hex screwdriver to remove the three screws on the bottom of Unit Cat1-CN, then flip and remove the bottom cover. The card slot location is shown in the red box below:

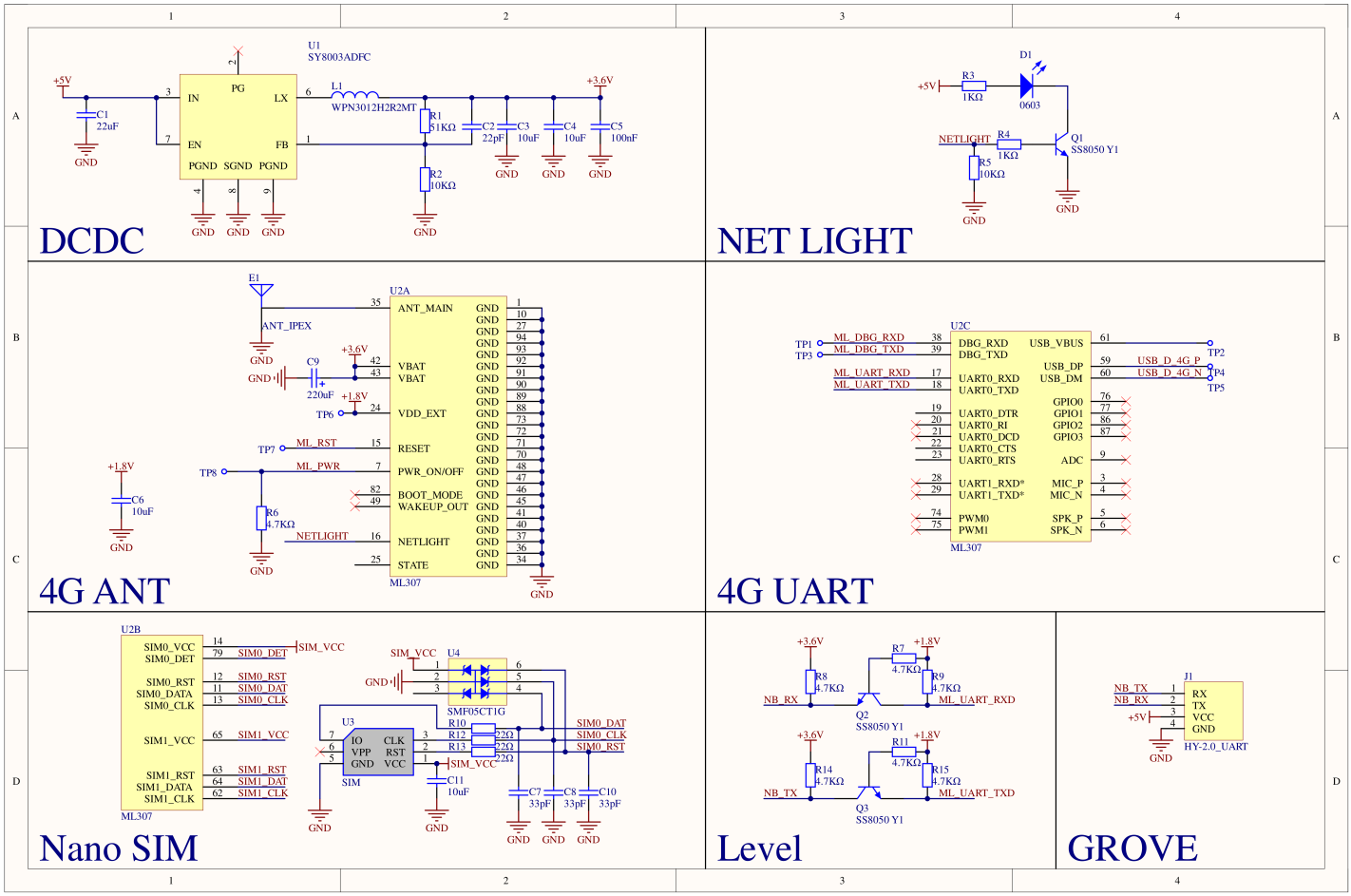


Baud Rate Limit

The recommended maximum UART baud rate for the Unit Cat1-CN module is **460800**. If configured to 921600, data loss or communication anomalies may occur due to the excessive speed; therefore, such configuration is not recommended.

Schematics

- Unit Cat1-CN Schematics PDF



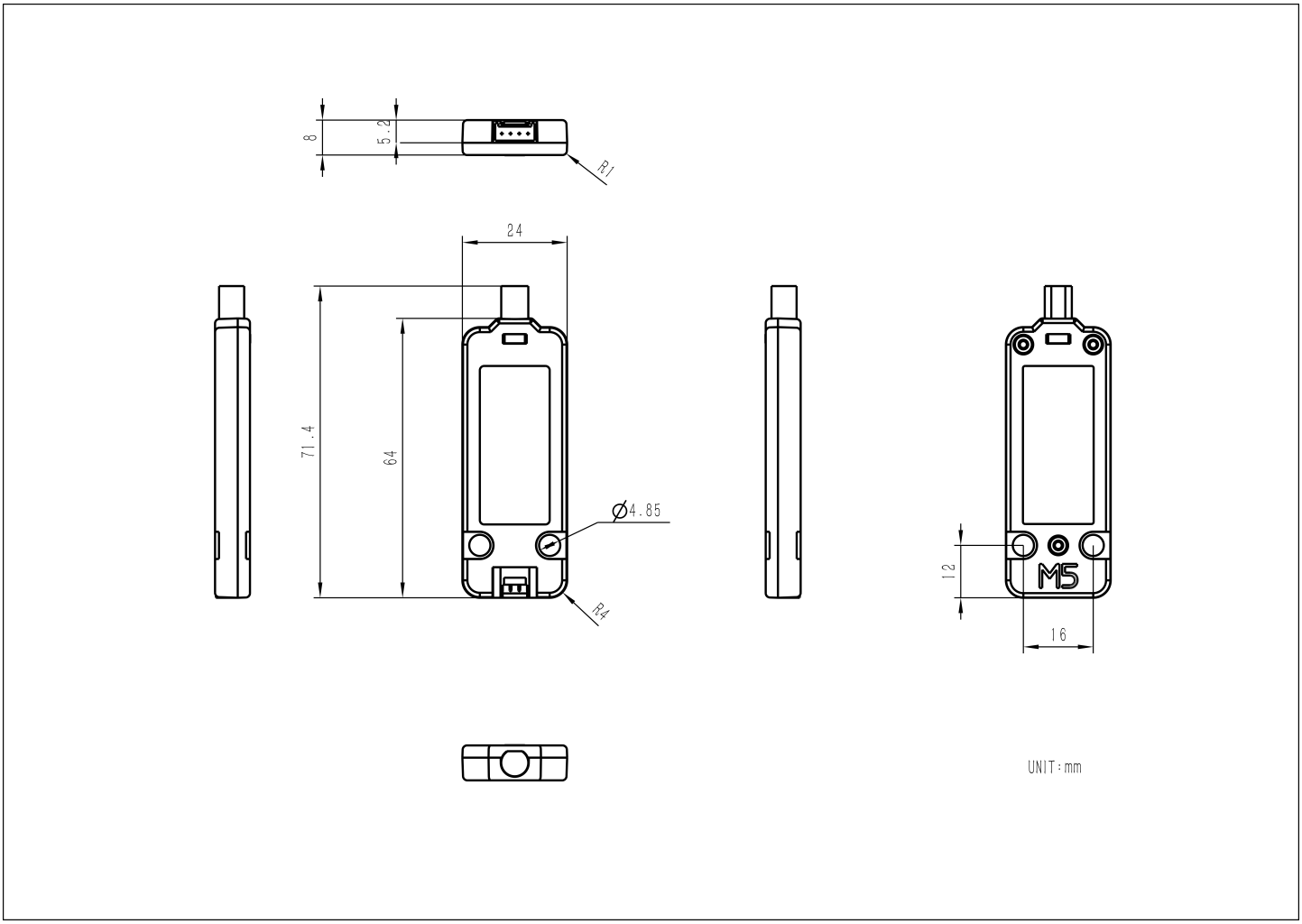
PinMap

Unit Cat1-CN

HY2.0-4P	Black	Red	Yellow	White
PORT.C	GND	5V	UART_RX	UART_TX

Model Size

- Unit Cat1-CN Model Size PDF



Structure

- Unit Cat1-CN Structure Files

Datasheets

- M307R Hardware Design Manual

Softwares

Arduino

- Unit Cat1-CN Arduino Quick Start
- Unit Cat1-CN Arduino Library

UiFlow2

- Unit Cat1CN UiFlow2 Docs

ESP-IDF

- ML307 ESP-IDF Component

Protocol

- ML307R AT Command Reference Manual
- ML307R TCP/IP User Manual
- ML307R SSL User Manual
- ML307R HTTPS User Manual
- ML307R MQTT User Manual
- ML307R Communication Process Example
- ML307R LBS User Manual
- ML307R LwM2M User Manual
- ML307R Dial-Up Internet Access User Manual
- ML307R Maintenance Diagnostic Tool User Manual
- ML307R Extended AT User Manual for 4G Series
- ML307R File System User Manual for 4G Series

Video

- Unit Cat1-CN Product Introduction and Function Demonstration

U204-Unit-Cat1-CN_video-EN.mp4